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Faculty of Science and Technology

Department of Computer Science

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Section: A

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Project Title

Outlay- Every Expense, Accounted For

Short Description

The Outlay – Every Expense, Accounted For is a role-based web application designed to manage and monitor work-related expenses incurred by employees during organizational activities. Employees may use their assigned budgets for expenses such as client-related activities, business trips, on-site medical issues, office supplies, and other approved work purposes. Employees are provided with a predefined budget before they incur expenses and can use the allocated amount when necessary.

After an expense occurs, the employee submits the expense to the appropriate superior for review. Managers review employee expenses and can approve or reject them based on the expense details and organizational requirements. If an expense is rejected, the employee is responsible for reimbursing the rejected amount to the company. Admins manage users, expense categories, budgets, and the approval of expenses submitted by Managers. The system records expense details, approval status, rejection reasons, budget usage, and remaining balances to provide better financial control, accountability, and transparency.

Introduction

Managing employee expenses manually through spreadsheets, paper records, receipts, or informal communication can make it difficult for an organization to monitor how employees use company-provided funds. Expenses may occur in different situations, such as travelling for business purposes, visiting or assisting clients, handling medical situations at a work site, purchasing office supplies, or performing other activities required for organizational operations. Without a centralized system, it can become difficult to determine which employee incurred an expense, what the expense was for, which budget was used, and whether the expense was approved.

The Outlay – Every Expense, Accounted For is proposed as a centralized expense and budget management system that focuses on the relationship between assigned employee budgets and actual work-related spending. Instead of providing employees with unrestricted access to company funds, the organization assigns a specific budget to employees in advance. Employees can then use their assigned budget for legitimate work-related expenses and submit the expenses after they occur.

The system follows a post-expense approval process. When an employee incurs an expense, they submit the expense details to their superior along with the relevant information required for review. The superior evaluates the expense and either approves or rejects it. An approved expense is recorded as an accepted organizational expense and reflected in the employee's budget usage. If the expense is rejected, the employee is responsible for reimbursing the rejected amount to the company. This creates a clear connection between the employee, assigned budget, actual expense, approval decision, and financial responsibility.

The system supports three roles: Admin, Manager, and Employee. Employees can view their assigned budgets, submit expenses, and monitor their expense status and remaining budget. Managers can review employee expenses, approve or reject submitted expenses, and monitor the budgets and expenses of their assigned employees. Admins can manage users, expense categories, budgets, and Manager expenses. Through this role-based structure, the system aims to improve expense accountability, budget visibility, and transparency within an organization.

Background Study

Existing Work

Existing expense-management systems provide a range of functions for recording, submitting, reviewing, approving, and monitoring organizational expenses. The reviewed systems show that employee expense submission and approval are established practices in modern expense-management software.

SAP Concur provides an integrated budget and expense-management environment in which organizations can configure budgets, track budget consumption, and provide budget visibility to budget owners, approvers, and viewers [1]. Expense reports can also be incorporated into configurable approval workflows, allowing organizations to control and review spending before it is finalized [2].

Zoho Expense provides similar expense-management functionality. Submitted transactions are routed to a user's assigned approver for review, and administrators can configure hierarchical approval flows so that expenses follow an organization's reporting structure automatically [3]. In addition, Zoho Expense explicitly supports budgets for individual users, allowing an administrator to assign a budget amount and

budget period to a particular employee [4]. This demonstrates that assigning budgets to individual employees is already an established feature in commercial expense-management software.

Ramp provides another example of an integrated expense and reimbursement platform. Employees can submit out-of-pocket expenses for reimbursement, after which the submitted reimbursement can be reviewed by the employee's direct manager or another configured reviewer [5]. Ramp also provides an employee-repayment mechanism for accidental or out-of-policy corporate-card transactions, allowing a reviewer to request full or partial repayment from the employee [6]. Therefore, employee financial responsibility for certain rejected or inappropriate transactions is also supported by an existing commercial system.

Expensify similarly provides an approval workflow in which submitted expense reports can be reviewed by assigned approvers. Approvers can approve or reject reports, while individual expenses can also be rejected with a reason and returned to the submitter for resolution and resubmission [7]. This demonstrates that post-expense submission, review, rejection, and resubmission are common features of existing expense-management systems.

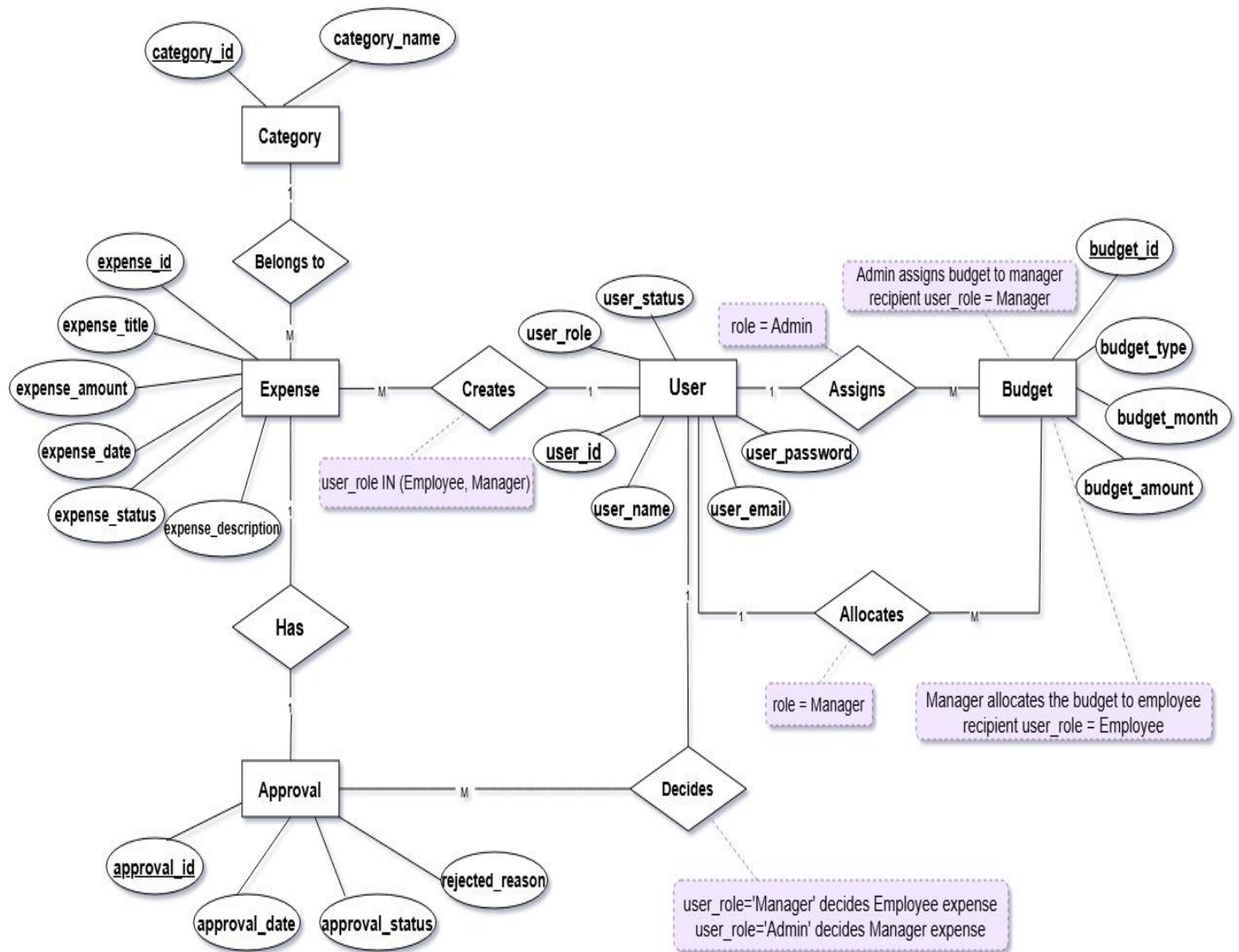
In Bangladesh, iBAS++ provides an example of a large-scale government financial management system. It is an Integrated Financial Management Information System used by the Government of Bangladesh and supports functions including budget preparation, budget execution, accounting, online bill submission and payment, and financial reporting [8]. iBAS++ also distributes approved budgets to offices and uses available-fund checking during financial transactions [8]. However, its scope is considerably broader than the employee-centered expense workflow addressed by the proposed system.

The reviewed systems demonstrate that individual functions of the proposed system such as employee expense submission, budgeting, hierarchical approval, rejection, and financial accountability already exist in different forms. Therefore, the proposed Outlay – Every Expense, Accounted For does not claim that these individual concepts are completely new. Instead, the proposed system integrates these concepts into a focused and lightweight organizational workflow.

The proposed system is specifically designed around employees who receive an assigned budget before incurring work-related expenses. Employees may use their assigned budget for activities such as taking care of clients, business trips, on-site medical situations, office supplies, and other organizational activities. After an expense occurs, the employee submits the expense for review by the appropriate superior. The superior can approve or reject the expense. If the expense is rejected, the proposed system records the rejected amount as an amount that the employee is responsible for reimbursing to the company.

Thus, the proposed work focuses on the integration of pre-assigned employee budgets, post-expense submission, superior approval or rejection, budget monitoring, and accountability for rejected expenses within a simplified Admin–Manager–Employee structure. The contribution of the project is therefore the design and implementation of this focused workflow rather than claiming that its individual functions have never been implemented by existing systems

ER Diagram



Budget Flow:

1. Admin assigns a total team budget to a Manager.
2. Manager receives his assigned portion given by admin and allocates the team budget portion to Employees.
3. Employees receives the allocated budget portion.

Figure 1: ER diagram

Normalization:

1. User → Creates → Expense:

Relationship: One User can create many Expenses (1:M).

UNF : user_id, user_name, user_email, user_password, user_role, user_status, expense_id, expense_title, expense_amount, expense_date, expense_status, expense_description

1NF : user_id, user_name, user_email, user_password, user_role, user_status, expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description

2NF:

Users: user_id (PK), user_name, user_email, user_password, user_role, user_status

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description, user_id (FK)

3NF:

Users: user_id (PK), user_name, user_email, user_password, user_role, user_status

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description, user_id (FK)

2. Category → Belongs to → Expense:

Relationship: One Category can contain many Expenses (1:M)

UNF: category_id, category_name, expense_id, expense_title, expense_amount, expense_date, expense_status, expense_description

1NF: category_id, category_name, expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description

2NF:

Category: category_id (PK), category_name.

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description, category_id (FK)

3NF:

Category: category_id (PK), category_name

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description, category_id (FK)

3. User → Decides → Approval:

Relationship: One User can decide many Approvals (1:M).

UNF: user_id, user_name, user_email, user_password, user_role, user_status, approval_id, approval_date, approval_status, rejected_reason

1NF: user_id, user_name, user_email, user_password, user_role, user_status, approval_id (PK), approval_date, approval_status, rejected_reason

2NF:

Users: user_id (PK), user_name, user_email, user_password, user_role, user_status

Approval: approval_id (PK), approval_date, approval_status, rejected_reason, user_id (FK)

3NF:

Users: user_id (PK), user_name, user_email, user_password, user_role, user_status

Approval: approval_id (PK), approval_date, approval_status, rejected_reason, user_id (FK)

4. Expense → Has → Approval:

Relationship: One Expense has one Approval (1:1).

UNF: expense_id, expense_title, expense_amount, expense_date, expense_status, expense_description, approval_id, approval_date, approval_status, rejected_reason

1NF: expense_id, expense_title, expense_amount, expense_date, expense_status, expense_description, approval_id (PK), approval_date, approval_status, rejected_reason

2NF:

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description

Approval: approval_id (PK), approval_date, approval_status, rejected_reason, expense_id (FK)

3NF:

Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description

Approval: approval_id (PK), approval_date, approval_status, rejected_reason, expense_id (FK UNIQUE)

***** expense_id is UNIQUE in Approval because one Expense can have only one Approval.**

5. User → Assigns → Budget:

Relationship: One Admin can assign many Budgets to Managers (1:M).

UNF: user_id, user_name, user_email, user_password, user_role, user_status, budget_id, budget_type, budget_month, budget_amount

1NF: user_id, user_name, user_email, user_password, user_role, user_status, budget_id (PK), budget_type, budget_month, budget_amount

2NF:

User: user_id (PK), user_name, user_email, user_password, user_role, user_status

Budget: budget_id (PK), budget_type, budget_month, budget_amount, assigned_by (FK)

3NF:

User: user_id (PK), user_name, user_email, user_password, user_role, user_status

Budget: budget_id (PK), budget_type, budget_month, budget_amount, assigned_by (FK), assigned_to (FK), parent_budget_id (FK NULL)

***** Role constraint:** assigned_by must refer to a User with user_role = Admin, and assigned_to must refer to a User with user_role = Manager for an Admin because Manager budget assignment.

6. User → Allocates → Budget:

Relationship: One Manager can allocate many Budgets to Employees (1:M).

UNF: user_id, user_name, user_email, user_password, user_role, user_status, budget_id, budget_type, budget_month, budget_amount

1NF: user_id, user_name, user_email, user_password, user_role, user_status, budget_id (PK), budget_type, budget_month, budget_amount

2NF:

User: user_id (PK), user_name, user_email, user_password, user_role, user_status

Budget: budget_id (PK), budget_type, budget_month, budget_amount, assigned_by (FK), assigned_to (FK), parent_budget_id (FK NULL)

3NF:

User: user_id (PK), user_name, user_email, user_password, user_role, user_status

Budget: budget_id (PK), budget_type, budget_month, budget_amount, assigned_by (FK), assigned_to (FK), parent_budget_id (FK NULL)

***** Role constraint:** assigned_by must refer to a User with user_role = Manager, and assigned_to must refer to a User with user_role = Employee for a Manager because Employee budget allocation.

***** Parent budget constraint:** parent_budget_id refers to the Manager's team budget from which the Employee's budget is allocated.

Finalization:

1. User: user_id (PK), user_name, user_email, user_password, user_role, user_status
2. Category: category_id (PK), category_name
3. Expense: expense_id (PK), expense_title, expense_amount, expense_date, expense_status, expense_description, user_id (FK), category_id (FK)
4. Approval: approval_id (PK), approval_date, approval_status, rejected_reason, user_id (FK), expense_id (FK UNIQUE)
5. Budget: budget_id PK, budget_type, budget_month, budget_amount, assigned_by (FK), assigned_to (FK), parent_budget_id (FK NULL)

Types of User: 1. User1 2. User2 3. User3

Common Features (Available to All Users)

- **Authentication**
 1. Login to the system
 2. Logout from the system
- **Account Management**
 1. Change or reset password
 2. Manage profile information (view, edit, delete)
- **Dashboard**
 1. Access a personalized dashboard after login

Specific Features by User Type

● **Features of User1: Admin**

1. User (Manager/Employee) Account Management

- View all registered users (Manager/Employee)
- Add/create user accounts with their assigned roles
- Edit user information
- Deactivate/delete user accounts

2. Expense Category Management

- Add new expense categories
- Edit category names
- Delete categories

3. System-Wide Expense Management

- View all users' expenses
- Search/filter expenses
- Monitor expense status
- Assign team budgets
- Modify assigned budgets when required
- Approve/reject manager expenses

● **Features of User2: Manager**

1. Review Employee Expense Requests

- View submitted employee expenses
- Open submitted expense details
- Check amount, category, date, and description

2. Approve or Reject & Monitor Expense Requests

- Approve valid expense requests
- Reject invalid expense requests
- Add rejection reason

- View employee expense summaries
- Monitor approved, rejected, and pending expenses

3. Manage Personal Budget Monitoring

- View personal monthly expense budget assigned by Admin
- Track personal expenses against the assigned budget
- View remaining available budget

4. Team Budget Oversight

- View the total team budget assigned by Admin
- View the remaining team budget available for employee allocation
- Assign individual budgets to employees within the available team budget
- Modify employee budgets within the available team allocation
- Monitor team spending

• Features of User3: Employee

1. Add Personal Expense

- Enter expense title
- Select category
- Enter amount, date and descriptions

2. Manage & Track Expenses

- View personal expenses and previous history
- Check expense status: Pending, Approved, or Rejected
- Edit/Delete pending expenses

3. Personal Expense Budget Monitoring

- View the monthly expense budget assigned by Manager
- Track personal expenses against the assigned budget
- View remaining available budget

Use Case Diagram

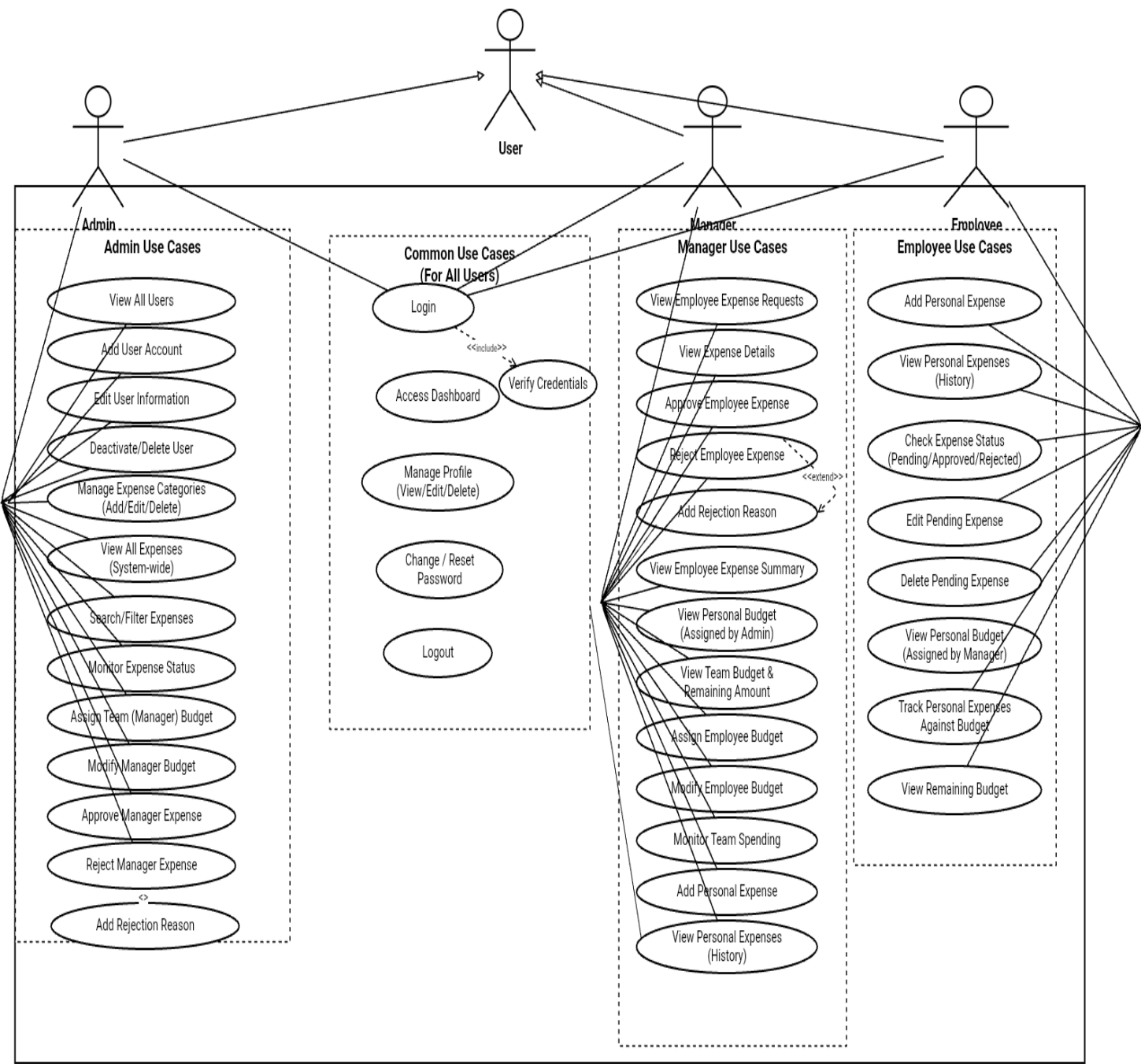


Figure: Use Case Diagram of Expense and Budget Management System

Figure 2: Use Case diagram

EXPENSE TRACKER

Welcome Back

Login to access your account

Email:

name@company.com

Password:

Enter your password

☐ Remember me

[Forgot Password ?](#)

Login

Figure 3: Login

EXPENSE TRACKER

Dashboard

Expenses

Budget

Categories

Users

Reports

My profile

Log Out

Welcome, Admin !

Total Users25

Total Expenses30,000 Tk

Approved18

Pending8

Date	User	Category	Amount	Status
2026-09-01	Rahim	Food	1000 Tk	Approved
2026-08-30	Karim	Travel	2000 Tk	Pending

Figure 4: Admin Dashboard

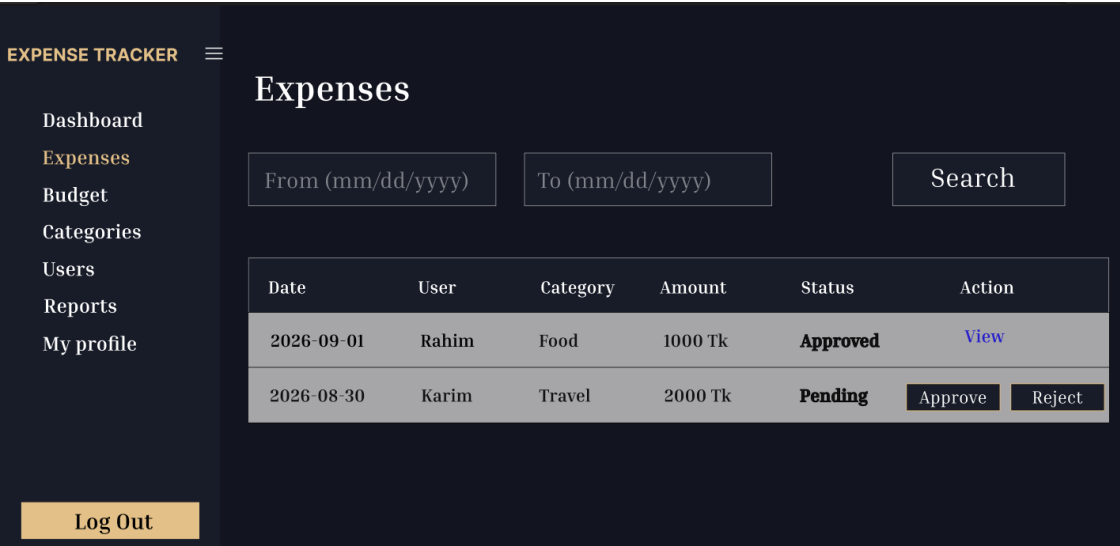


Figure 5: Admin Expense

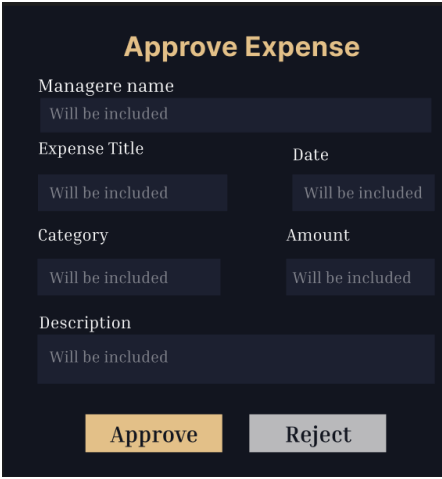


Figure 6: Admin Expense Approve

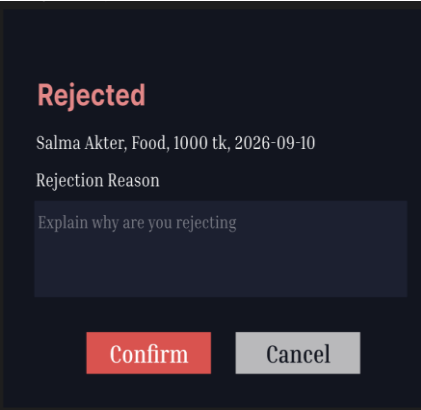


Figure 7: Admin Expense Rejection



Figure 8: Admin Budget

Assign Budget

Manager:

Select a manager

Budget amount:

Enter an amount

Budget Date:

mm/dd/yyyy

Budget Type:

Select a budget type

Assign

Cancel

Figure 9: Assign Budget

EXPENSE TRACKER

Dashboard

Expenses

Budget

Categories

Users

Reports

My profile

Log Out

Expense Categories

+ Add Category

ID	Category Name	Action
1	Food	
2	Travel	
3	Office	
4	Utilities	

Figure 10: Admin Category

Add category

Category Name:

Description

Enter Description (Optional)

Assign

Cancel

Figure 11: Add Category

Edit category

Category Name:

Edit Category Name:

Edit

Cancel

Figure 12: Edit Category

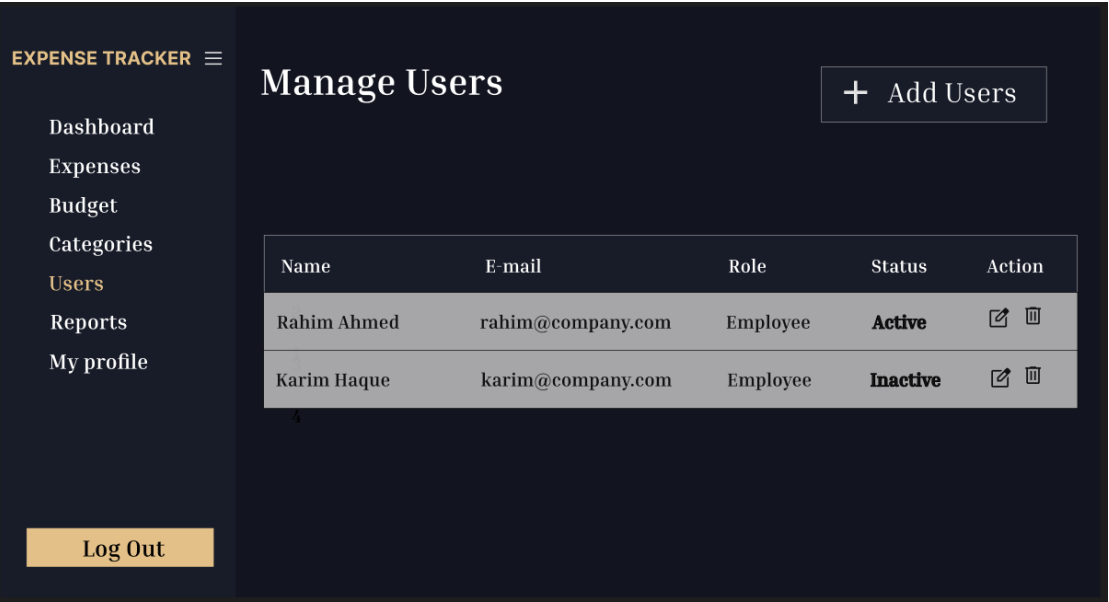


Figure 13: Admin Users

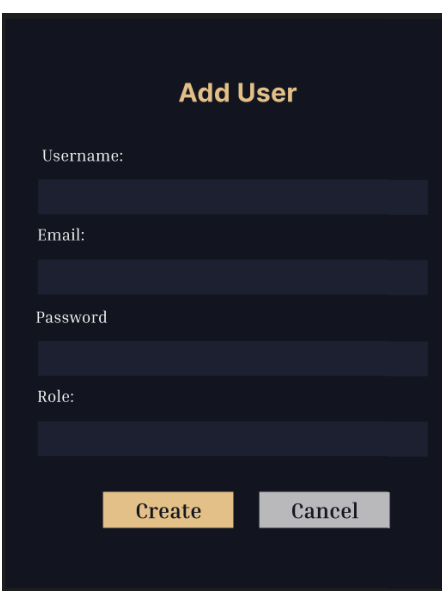


Figure 14: Add User

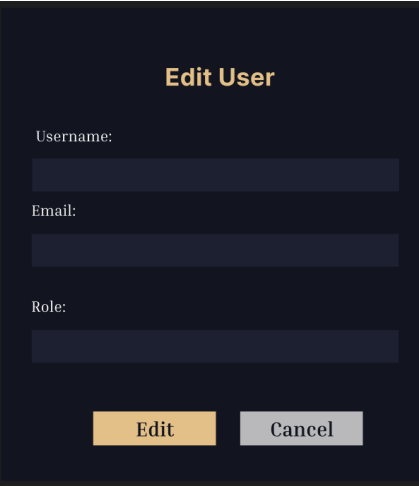


Figure 15: Edit User

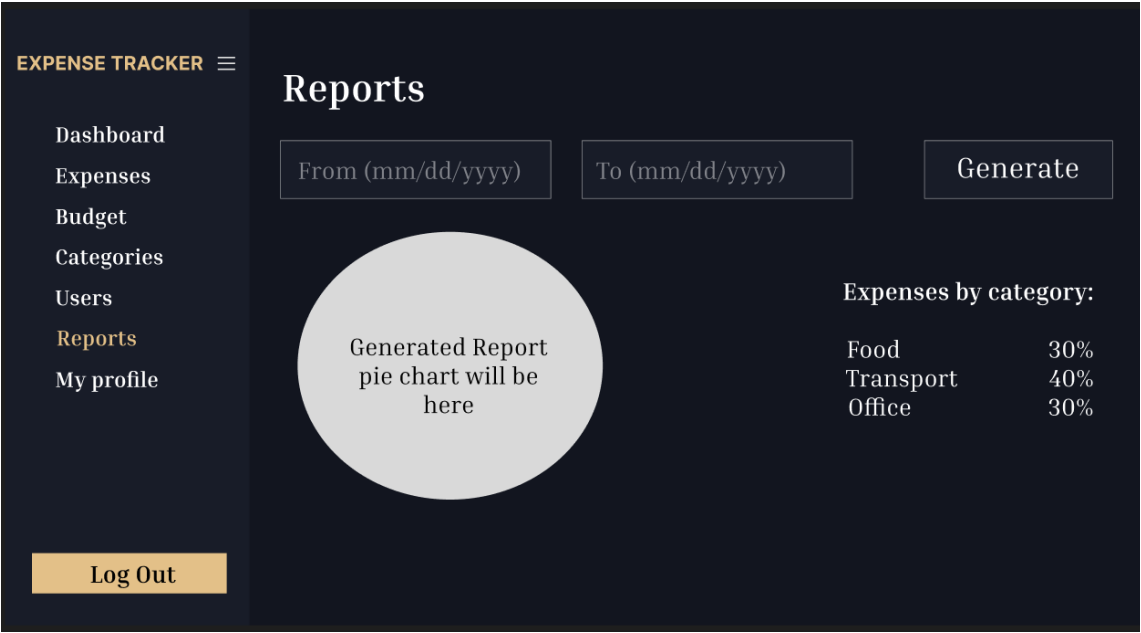


Figure 16: Reports

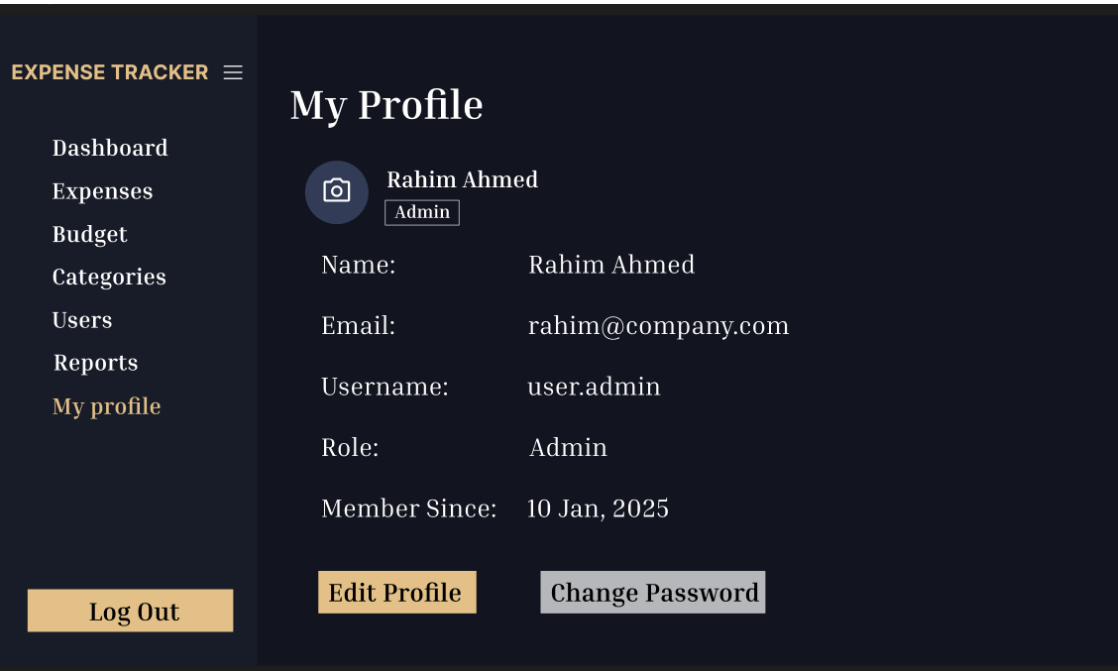


Figure 17: My Profile

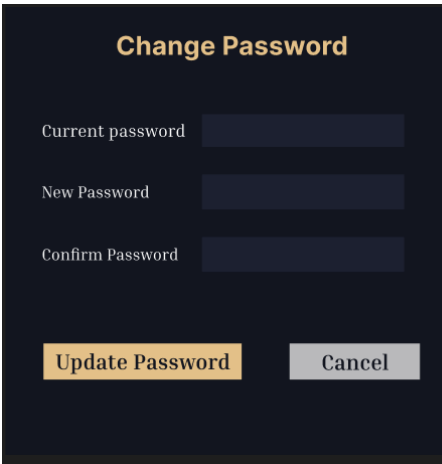


Figure 18: Change Password



Figure 19: Manager Dashboard



Figure 20: Manager monitoring employee expenses

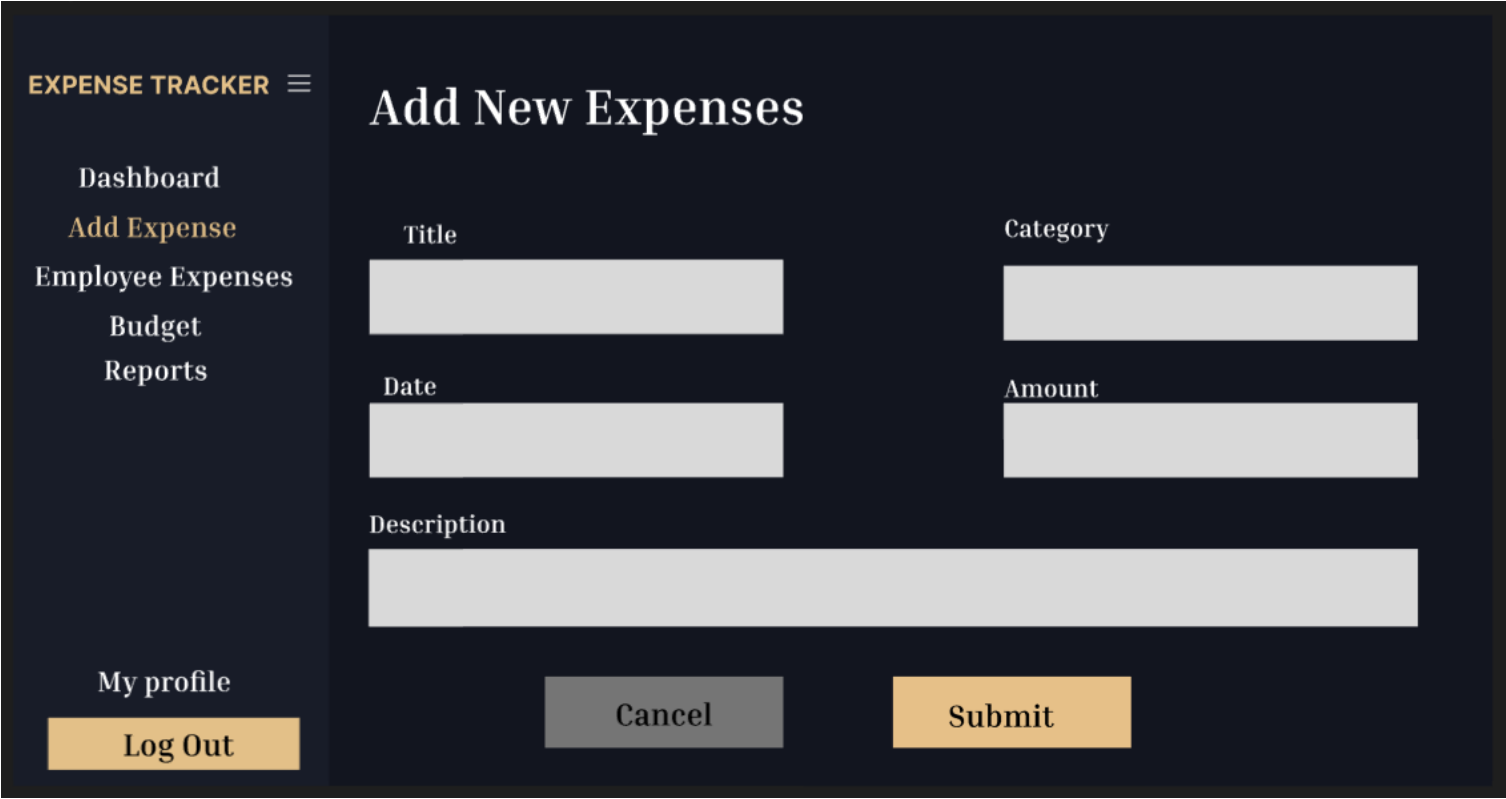


Figure 21: Add Expense



Figure 22: Manager Expense



Figure 23: Expense Details

EXPENSE TRACKER

Dashboard

Add Expense

Employee Expenses

Budget

Reports

My profile

Log Out

Approve Expense

Employee name

Will be included

Expense Title

Will be included

Date

Will be included

Category

Will be included

Amount

Will be included

Description

Will be included

Approve

Reject

Figure 24: Manager Approvals

EXPENSE TRACKER

Dashboard

Add Expense

Employee Expenses

Budget

Reports

My profile

Log Out

My Budget

Allocated

10,000

Spent

2000

Remaining

8000

Team Budget

Allocated

10,000

Spent

2000

Remaining

8000

Category	Amount	% of team Budget
Food	1000 Tk	20%
Travel	2000 Tk	40%

Figure 25: Manager Budgets (My Budget, Team Budget)

EXPENSE TRACKER

Dashboard

Add Expense

My Expenses

Budget

Reports

My Profile

Log Out

Good Morning, Fatima!

My budget

20,000 Tk

Spent

12, 000 Tk

Pending

1

Remaining

7,500 Tk

Add Expense

Recent expenses

Title	Category	Amount	Status
Lunch with team	Food	850 Tk	Approved
Client site visit	Travel	2500 Tk	Pending

Figure 26: Employee Dashboard



Figure 27:Employee expense

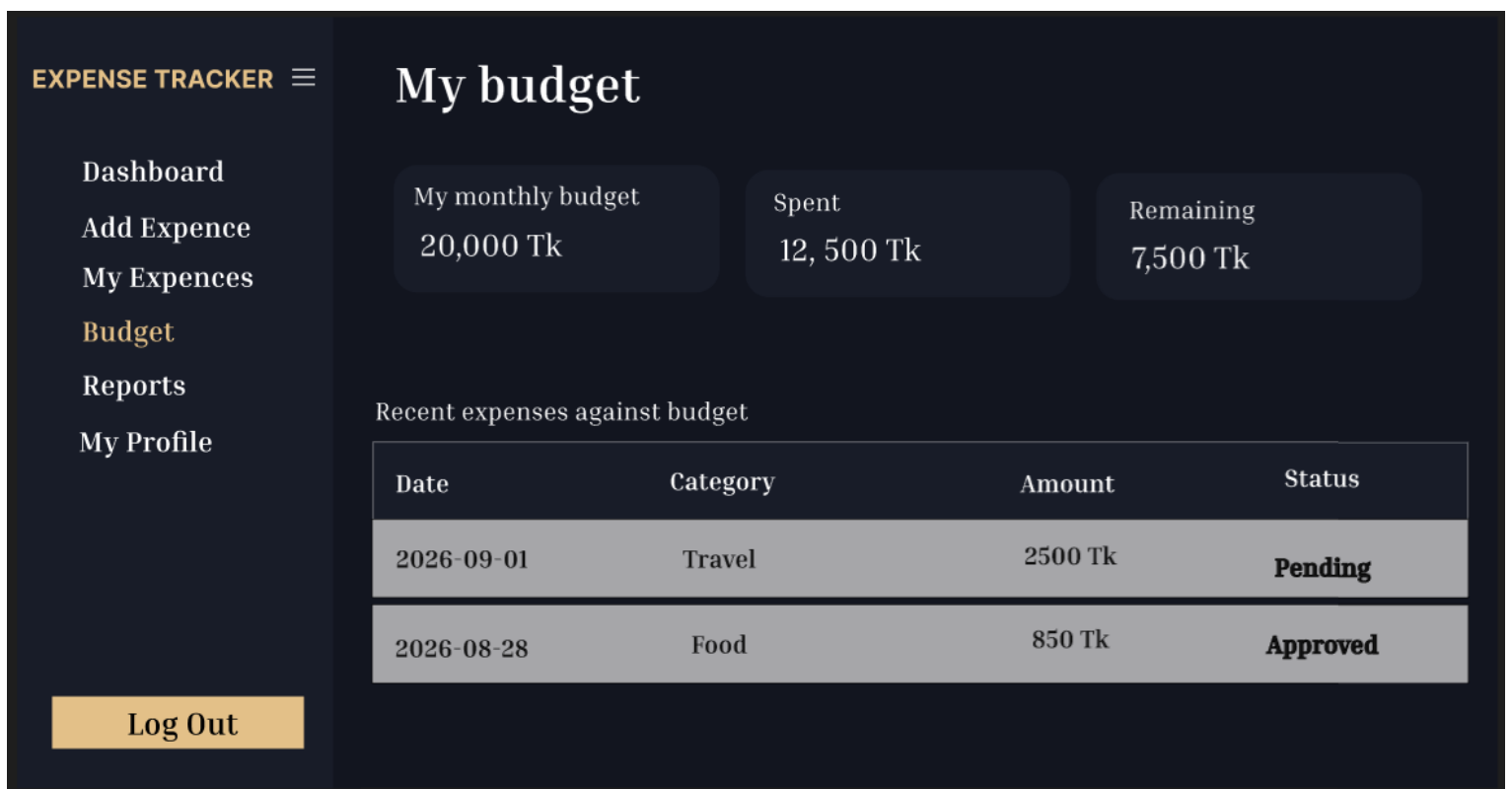


Figure 28: Employee Budget

References

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